ADHWAIDE KRISHNA

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PROFESSIONAL SUMMARY

AI Engineer | RAG, Agentic Systems & Applied LLMs

AI & Machine Learning graduate who builds end-to-end LLM systems: RAG pipelines, multi-agent research platforms, knowledge graphs, and CPU-only computer vision. Four full-stack projects in Python, FastAPI, and React using Gemini and open-source LLMs, each designed to be explainable, measurable, and runnable on modest hardware.

TECHNICAL SKILLS

AI/ML Techniques: RAG, Vector Databases, Semantic Search, Agentic AI, Multi-Agent Systems, Prompt Engineering
AI Systems & LLMs: Ollama, Hugging Face, Google AI Studio, Mistral, Llama, Qwen, Gemma, Local LLM Deployment
Programming & Frameworks: Python, FastAPI, React, REST APIs, WebSockets, Django
ML Frameworks: YOLOv8, TensorFlow, PyTorch, NLP, Computer Vision, Object Detection, Speech Recognition, Text-to-Speech
Data & Tools: Pandas, NumPy, Power BI, Tableau, Git, GitHub, Google Colab, AWS (Familiar)

AI / ML PROJECTS

Digital Immortality Platform – AI Persona Simulation System | Lead Developer *Mar 2025 – Present*
Tech: Python, FastAPI, React, SQLite, Gemini API, NumPy |

- Architected a RAG-based digital persona system with sentence-aware chunking (800 chars, 100 overlap) and 768-dim Gemini embeddings, using separate document and query embedding modes for context-aware replies.
- Built a custom NumPy vector memory with category and mood filtering and a 0.4 similarity gate, keeping retrieval under 10 ms and end-to-end replies at 0.8–2.0 s.
- Enforced persona consistency with a layered prompt (Big Five traits, ranked values, 10 scenario anchors from a 16-question quiz) and a per-reply grounding confidence score based on top-memory similarity.
- Implemented an autonomous memory reflection pipeline that synthesizes and stores new episodic, semantic, and behavioral memories from interaction history.

Autonomous Research Analyst (ARA) *Jun 2026 – Jul 2026*
Tech: Python, Ollama (Llama 3.2), Sentence-Transformers, SQLite, NetworkX, RAG, Knowledge Graphs |

- Built an 8-agent research platform that ingests up to 23 documents per cycle from arXiv, RSS, and GitHub, deduplicates them by URL hash, and extracts entities and relations into a knowledge graph using schema-constrained LLM prompts.
- Implemented RAG with all-MiniLM-L6-v2 embeddings (384-dim) and 1,200-char chunking with 200 overlap, plus hybrid retrieval that combines the top-4 vector chunks with knowledge-graph facts in the prompt.
- Built reasoning agents for hypothesis generation (open-triad graph analysis), 3-level causal what-if trees, and a 6-persona debate council whose consensus claims are written back to the graph.

CatalystAI – Stock Intelligence & Research Platform *May 2026 – Jun 2026*
Tech: Python, Ollama, NetworkX, SQLite, Knowledge Graphs, Vector Embeddings | GitHub: github.com/Adhwaide/CatalystAI

- Built a knowledge-graph stock intelligence platform with 220 nodes and 330 relationships, covering 115 Indian listed companies across 20 sectors, with BFS-based causal traversal in under 5 ms.
- Engineered a multi-factor scoring engine (relationship strength, revenue exposure, source reliability, market validation) with local LLM entity extraction and explainable theses in 10–25 s per event, and a deterministic fallback under 100 ms when Ollama is offline.
- Built a backtesting framework that validates rankings on 100 historical events (2023–2025) against Nifty 50, sector, and random-pick baselines, tracking Precision@k and 30-day excess return across 16,800+ stored evaluations.

V1B3 Cognitive Memory System *Apr 2026 – Jun 2026*
Tech: Python, FastAPI, React, SQLite, OpenCV, YOLOv8 |

- Engineered a five-layer cognitive memory architecture (perception, episodic, semantic, spatial, procedural) for persistent object tracking and retrieval.
- Developed a custom 1280-dim visual embedding (768-dim spatial color grid + 512-bin HSV histogram, L2-normalized) for object re-identification with sub-0.1 ms matching, running at ~15–22 FPS on an Intel i3 CPU with no GPU.
- Built an explainable reasoning dashboard with knowledge graph visualization, memory replay, and step-by-step decision tracing.

PROFESSIONAL EXPERIENCE

Python Developer Intern | Techmindz, Kochi *Dec 2024*

- Built a full-stack Django web application with integrated frontend, backend, authentication, and database optimization.
- Developed RESTful API endpoints enabling efficient and secure client-server communication.

EDUCATION

B.Sc. in Artificial Intelligence & Machine Learning *May 2026*
CMS College of Science and Commerce, Coimbatore, Tamil Nadu
Relevant Coursework: Data Structures, Python, C++, Business Data Analytics

CERTIFICATIONS & ADDITIONAL

Certifications: Data Visualisation – TATA Forage
Languages: English (Fluent) | Hindi (Fluent) | Tamil (Fluent) | Malayalam (Native)